SAUNG HNINN PHYU

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PROFESSIONAL SUMMARY

Full Stack and AI Engineer with hands-on experience building full-stack, AI-powered applications including multi-agent LLM systems, RAG pipelines, and production-grade SaaS platforms. Fresh graduate at the University of Wollongong with a track record of shipping 8+ end-to-end projects using React/Next.js, Python, and Azure/OpenAI APIs. Award-winning hackathon builder (Community Choice, Best Use of Microsoft Stack).

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript, TypeScript
- **AI & LLM:** OpenAI API, Azure OpenAI, Azure AI Foundry, RAG (Retrieval-Augmented Generation), Multi-Agent Systems, Prompt Engineering, Hugging Face Transformers, LLM Fine-Tuning
- **Web Development:** React, Next.js, HTML, CSS, TailwindCSS
- **Backend & APIs:** Node.js, Express.js, REST API Development, RESTful Services
- **Databases:** PostgreSQL, Supabase (Row-Level Security, Auth), Prisma, SQL Server, Firebase
- **Machine Learning & Data:** Scikit-learn, PySpark, Feature Engineering, Cross-Validation, Model Evaluation (BLEU, chrF)
- **Auth & Identity:** Microsoft Entra ID, Auth.js
- **Version Control & DevOps:** Git, GitHub, CI/CD Pipelines, Vercel Deployments
- **Testing:** Jest (Unit Testing), Playwright (E2E Testing)
- **Methodologies:** Agile Scrum

PROJECTS

Smart Task Allocation | *Next.js 14, Supabase (PostgreSQL, Auth, RLS), Tailwind CSS, OpenAI, Azure OpenAI, Vercel*

- Independently built a multi-tenant SaaS platform that helps cleaning companies assign staff to client jobs, isolating each company's data with company-scoped Supabase Row-Level Security across 6 user roles.
- Replaced manual staff selection with an AI recommendation engine that ranks workers by availability, proximity, workload and performance.
- Cut weekly schedule creation from ~20 minutes to under 2 minutes by building a natural-language scheduling agent (Azure OpenAI) that turns chat prompts into recurring contracts and schedules, with role-based checks that block unauthorised staff from scheduling actions.
- Automated daily operations with QR-based attendance, cron-driven recurring bookings and an AI-assisted customer booking wizard.
- Built a shared component library reused across all 6 role-based modules, keeping the UI consistent and reducing duplicated code.
- Wrote 84 tests (48 Jest unit, 36 Playwright E2E) covering the recommendation engine, booking logic and cross-tenant access control; deployed on Vercel.

Emochi - AI Emotional Wellness Platform | [Hackxperience, 2026] | *Next.js, Azure AI Foundry, Prisma/SQL Server, Auth.js*

- Built the AI agent layer of a wellness chat app where users get emotional support from 8 specialised personas; the [4]-person team won Community Choice and Best Use of Microsoft Stack out of [14] teams.
- Configured 8 LLM agents in Azure AI Foundry, each grounded in its own RAG knowledge base, replacing generic chatbot replies with persona-specific responses.
- Delivered real-time chat by integrating the agents through the OpenAI-compatible Responses API with token-level streaming.
- Designed a multi-agent debate mechanic where a "director" agent selects relevant personas, enforces genuine disagreement and issues a judged verdict.
- Wrote agent behaviour guidelines to keep responses safe and non-harmful in a sensitive wellness context, and to define each agent's role in multi-agent conversations.

Online Fundraising System | *TypeScript, React, Supabase, GitHub, Agile Scrum*

- Co-developed a web-based fundraising platform where fundraisers create and manage campaigns, donees browse, save and track campaigns, and platform managers oversee categories and generate daily/weekly/monthly reports.

- Designed and built the Admin user and profile management module, from wireframes and UI/UX flows through to implementation using a Boundary-Controller-Entity architecture with Supabase.
- Modeled the system's structure and key interactions through a class diagram and sequence diagrams, giving the dev team a clear technical blueprint.
- Authored the ethical considerations section, surfacing the platform's data and trust concerns before development began.
- Wrote the test plan and ran UI/UX testing to validate designs against requirements across 5 Agile Scrum sprints, collaborating via GitHub and Taiga.

AI Customer Support Chatbot | *JavaScript, React, Express.js, Supabase, OpenRouter API, CSS*

- Designed and built a full-stack AI chatbot that answers customer queries for [an online store], pairing a React frontend with an Express.js REST API and Supabase for persistence.
- Integrated the OpenRouter API to generate chatbot responses and added persistent conversation history so users can pick up past chats.
- Built in-chat order tracking so customers can check their order status directly, resolving one of the most common support queries without a human agent.

Neural Machine Translation System | *Python, Hugging Face Transformers, MarianMT, mBART-50, NLLB-200*

- Benchmarked 3 transformer translation models across 6 translation directions (English ↔ Indonesian, Vietnamese, Korean), with each model fine-tuned on 10,000 OPUS-100 sentence pairs per language pair.
- Kept fine-tuning of the 600M-parameter multilingual models running within limited GPU memory by applying gradient accumulation, small batch sizes and mixed-precision (fp16) training.
- Visualised and analysed BLEU and chrF results before and after fine-tuning, showing that the smaller bilingual MarianMT outperformed the larger multilingual models in every direction while also running fastest.

Network Intrusion Detection System | *Python, Scikit-learn, PySpark*

- Built a Spark MLlib pipeline to classify 175,000+ UNSW-NB15 network traffic records as normal or attack.
- Trained and compared 4 classifiers: Logistic Regression, Decision Tree, Random Forest and Naive Bayes.
- Applied data preprocessing and correlation-based feature selection (42 → 21 features) and engineered traffic interaction features with RFormula.
- Improved model reliability with 3-fold cross-validation and hyperparameter tuning of the best-performing Decision Tree.

EDUCATION

University of Wollongong, Singapore Oct 2024 – Sep 2026
Bachelor of Computer Science, Major in Big Data

Singapore Institute of Management Oct 2023 – Sep 2024
Diploma in Information Technology, GPA 3.63/4.00

AWARDS & CERTIFICATIONS

- AWS Certified Cloud Practitioner Oct 2026
- Community Choice Award & Best Use of Microsoft Stack - Emochi (Hackathon)
- Python for Data Science Skill Path
- AgentCamp 2026 - Global AI Singapore, Certificate of Participation

REFERENCE

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